POLY PMNA · DIPLOMA CURRICULUM REV2021

Course 2139 | Semester 2 | Lab | Shared handbook

Problem Solving and Programming Lab

A practical laboratory handbook that converts programming concepts into repeatable experiments, records, debugging exercises and a small evidence-based mini-project.

Course code	**2139**
Revision	**REV2021**
Semester	**Semester 2**
Type	**Lab**
Catalogue scope	**9 catalogue programme assignments**

COURSE ORIENTATION

How to use this handbook

This page is a structured learning aid for the shared catalogue record. Study the concept, complete the applied activity, retain the evidence, and compare the result with the latest approved programme instructions. Where the official syllabus differs, the official record takes precedence.

Source and verification note: The course identity, code, semester and subject type are taken from the tracked POLY PMNA REV2021 catalogue, which indexes the official SITTTR scheme. The official course-content reference is SITTTR course 2139; the scheme index is SITTTR REV2021 diploma syllabus. The direct subject-content page was not machine-readable or returned an unavailable-file response during preparation, so module headings and explanatory teaching material on this page are an original study-handbook structure, not claimed verbatim SITTTR wording. Confirm current hours, marks, credits, outcomes and assessment against the latest official programme record before examination use.

Learning outcomes

- Explain the central concepts and vocabulary of the subject using clear technical language.
- Translate a problem into an algorithm, implementation or practical experiment and test it with meaningful cases.
- Create readable code or a laboratory record with algorithm, output, test evidence and reflection.
- Identify assumptions, limitations and common errors, then improve the work using feedback.

Shared-record context

This code is assigned to 9 catalogue programme assignments in the tracked catalogue. The page therefore focuses on common learning practice and avoids claiming programme-specific technical details that were not verified.

Study rule: Do not treat the author-created examples as official examination questions or official mark distribution.

STUDY MAP

Modules and evidence

Author-created study map; verify official hours and marks			
Module	**Heading**	**Purpose**	**Evidence to retain**
1	Laboratory orientation and records	Set up a safe, reproducible programming workspace and maintain a complete lab record.	Create the first lab record from a simple input/output experiment.
2	Core programming experiments	Implement and test a sequence of small programs that build confidence in syntax and control flow.	Complete four experiments and attach at least three test cases to each one.
3	Functions, strings and collections	Use modular code and structured data in practical problems.	Write a reusable function library for two related lab exercises.
4	Debugging and verification practicals	Find, explain and correct common faults without changing the problem requirement.	Diagnose a deliberately faulty program and document the evidence for each correction.
5	Mini-project and viva	Integrate the laboratory skills into a small programme-relevant solution and defend the implementation.	Submit a mini-project folder with source, record, test evidence and a two-minute viva script.

MODULE 1

Laboratory orientation and records

Purpose-led study | Apply + reflect

Set up a safe, reproducible programming workspace and maintain a complete lab record.

Core topics–

- Workstation and file safety
- Naming, folders and backups
- Academic integrity and responsible reuse
- Experiment record: aim, algorithm, code, output, result and reflection

Study method

Read the concept, connect it to a diploma-level situation, make a labelled record or diagram, and check the result against the stated outcome. Keep programme-specific technical decisions within the approved project, laboratory or workplace context.

Applied activity

Create the first lab record from a simple input/output experiment.

Evidence: Keep the completed activity, a short reflection and any source or test record in your course folder.

Common mistakes and corrections−

- Starting with implementation before defining the problem, inputs, outputs or acceptance criteria.
- Making a claim without a source, measurement, test or documented observation.
- Using a template mechanically instead of adapting it to the approved programme context.

Correction: state the assumption, choose evidence, perform the check, and record the decision.

Self-check−

1. Explain the main idea of this module in your own words.
2. Apply it to a small technical or workplace scenario.
3. Identify one likely failure and describe the evidence that would diagnose it.

MODULE 2

Core programming experiments

Purpose-led study | Apply + reflect

Implement and test a sequence of small programs that build confidence in syntax and control flow.

Core topics−

- Input/output and arithmetic
- Unit conversion and formatted output
- Selection and validation
- Loops, counters and nested loops

Study method

Read the concept, connect it to a diploma-level situation, make a labelled record or diagram, and check the result against the stated outcome. Keep programme-specific technical decisions within the approved project, laboratory or workplace context.

Applied activity

Complete four experiments and attach at least three test cases to each one.

Evidence: Keep the completed activity, a short reflection and any source or test record in your course folder.

Common mistakes and corrections−

- Starting with implementation before defining the problem, inputs, outputs or acceptance criteria.
- Making a claim without a source, measurement, test or documented observation.
- Using a template mechanically instead of adapting it to the approved programme context.

Correction: state the assumption, choose evidence, perform the check, and record the decision.

Self-check−

1. Explain the main idea of this module in your own words.
2. Apply it to a small technical or workplace scenario.
3. Identify one likely failure and describe the evidence that would diagnose it.

MODULE 3

Functions, strings and collections

Purpose-led study | Apply + reflect

Use modular code and structured data in practical problems.

Core topics−

- Functions and parameter passing

- String traversal and validation
- Collection traversal, search and sort
- Output formatting and evidence capture

Study method

Read the concept, connect it to a diploma-level situation, make a labelled record or diagram, and check the result against the stated outcome. Keep programme-specific technical decisions within the approved project, laboratory or workplace context.

Applied activity

Write a reusable function library for two related lab exercises.

> **Evidence:** Keep the completed activity, a short reflection and any source or test record in your course folder.

Common mistakes and corrections−

- Starting with implementation before defining the problem, inputs, outputs or acceptance criteria.
- Making a claim without a source, measurement, test or documented observation.
- Using a template mechanically instead of adapting it to the approved programme context.

Correction: state the assumption, choose evidence, perform the check, and record the decision.

Self-check−

1. Explain the main idea of this module in your own words.
2. Apply it to a small technical or workplace scenario.
3. Identify one likely failure and describe the evidence that would diagnose it.

MODULE 4

Debugging and verification practicals

Purpose-led study Apply + reflect

Find, explain and correct common faults without changing the problem requirement.

Core topics−

- Syntax, runtime and logical errors
- Boundary and invalid-input tests
- Assertions and regression checks
- Defect log and corrected output

Study method

Read the concept, connect it to a diploma-level situation, make a labelled record or diagram, and check the result against the stated outcome. Keep programme-specific technical decisions within the approved project, laboratory or workplace context.

Applied activity

Diagnose a deliberately faulty program and document the evidence for each correction.

Evidence: Keep the completed activity, a short reflection and any source or test record in your course folder.

Common mistakes and corrections−

- Starting with implementation before defining the problem, inputs, outputs or acceptance criteria.
- Making a claim without a source, measurement, test or documented observation.
- Using a template mechanically instead of adapting it to the approved programme context.

Correction: state the assumption, choose evidence, perform the check, and record the decision.

Self-check−

1. Explain the main idea of this module in your own words.
2. Apply it to a small technical or workplace scenario.
3. Identify one likely failure and describe the evidence that would diagnose it.

MODULE 5

Mini-project and viva

Purpose-led study | Apply + reflect

Integrate the laboratory skills into a small programme-relevant solution and defend the implementation.

Core topics−

- Requirements and algorithm
- Incremental implementation
- Test plan and demonstration
- Lab record, reflection and viva

Study method

Read the concept, connect it to a diploma-level situation, make a labelled record or diagram, and check the result against the stated outcome. Keep programme-specific technical decisions within the approved project, laboratory or workplace context.

Applied activity

Submit a mini-project folder with source, record, test evidence and a two-minute viva script.

Evidence: Keep the completed activity, a short reflection and any source or test record in your course folder.

Common mistakes and corrections−

- Starting with implementation before defining the problem, inputs, outputs or acceptance criteria.
- Making a claim without a source, measurement, test or documented observation.
- Using a template mechanically instead of adapting it to the approved programme context.

Correction: state the assumption, choose evidence, perform the check, and record the decision.

Self-check−

1. Explain the main idea of this module in your own words.
2. Apply it to a small technical or workplace scenario.
3. Identify one likely failure and describe the evidence that would diagnose it.

ASSESSMENT PREPARATION

Evidence, revision and quality gates

Before submission or examination

1. Confirm the current SITTTR/SBTE title, hours, credits, outcomes and assessment.
2. Define the problem or prompt and state assumptions.
3. Show the method, calculation, algorithm, project record or communication structure.
4. Attach test, source, supervisor, workplace or reflection evidence as applicable.
5. Review clarity, safety, ethics, citations, originality and accessibility.

Revision checklist

- Can I define the main terms without copying?
- Can I apply the method to a new situation?
- Can I explain one common error and its correction?
- Can I show evidence for each claimed outcome?
- Can I state the limitations and the next improvement?

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Evidence record
Subject: 2139 — Problem Solving and Programming Lab
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Task or question:
Assumption or requirement:
Method / design / procedure:
Result or artefact:
Test / source / supervisor evidence:
Reflection and next action:
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SELF-CHECK AND EXAMINATION PRACTICE

Question bank

Recall question

Define two important terms from Problem Solving and Programming Lab and distinguish them.
Answer guidance: define the terms, state assumptions, show the method, use a specific example, and cite or retain evidence where appropriate.

Explain question

Explain why the method in this handbook matters for a diploma student.
Answer guidance: define the terms, state assumptions, show the method, use a specific example, and cite or retain evidence where appropriate.

Apply question

Apply one module method to a realistic technical, project or workplace situation.
Answer guidance: define the terms, state assumptions, show the method, use a specific example, and cite or retain evidence where appropriate.

Evaluate question

Identify a weak approach, state the evidence needed, and propose a defensible improvement.
Answer guidance: define the terms, state assumptions, show the method, use a specific example, and cite or retain evidence where appropriate.

Create question

Design a small artefact, plan, test, report section or presentation that demonstrates the subject outcomes.

Answer guidance: define the terms, state assumptions, show the method, use a specific example, and cite or retain evidence where appropriate.

REFERENCES AND GLOSSARY

Reference trail

1. Official SITTTR REV2021 diploma syllabus catalogue.
2. Official SITTTR course-content reference for code 2139.
3. POLY PMNA Study Hub, for the current revision index and related lesson pages.

Glossary

Terms used in this handbook

Term	Working meaning
Outcome	A capability the learner should demonstrate, not just a topic read.
Evidence	A record, artefact, result, source, test or review that supports a claim.
Acceptance criterion	A condition used to decide whether a requirement or deliverable is satisfactory.
Reflection	A brief evidence-based account of what worked, what did not and what should change.

Prepared as a POLY PMNA study handbook. Always follow the latest official SITTTR/SBTE instructions for the programme and examination session.